

# Ankit Rao

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## EDUCATION

### University of Virginia

Charlottesville, VA

*B.A. in Computer Science (GPA: 3.77/4.0)*

*May 2028*

- Coursework: Data Structures & Algorithms, Computer Systems and Architecture, Discrete Mathematics, Linear Algebra, AI-Powered Cyber Security, Software Development Essentials, Deep Learning, Databases, Artificial Intelligence, Probability
- Honors: Coca-Cola Scholar Semifinalist (top 1.27%), MIT Research Conference Accepted Presenter

## EXPERIENCE

### Scyntic | [scyntic.ai](#)

Apr. 2026 – Present

*Co-Founder and CTO*

*Charlottesville, VA*

- Reduced cost per agent run by 84% and analysis runtime by 87% by optimizing execution pipelines via PgBouncer connection pooling and async task batching
- Scaled concurrent agent capacity to 1,000+ simultaneous runs by architecting AWS ECS infrastructure using Redis and Celery for parallel task distribution across merchant storefronts
- Enabled merchants to identify and fix checkout drop-off in real time by building reporting pipelines for A/B tests, telemetry aggregation, and code recommendations
- Automated end-to-end storefront testing for ecommerce merchants by building an AI agent platform that sends simulated shoppers through stores to surface conversion friction points

### Howard Hughes Medical Institute

June 2025 – Aug. 2025

*Machine Learning Intern*

*Ashburn, VA*

- Improved 3D cell-tracking accuracy by 30% on terabyte-scale imaging datasets by deploying a CUDA-accelerated PyTorch pipeline on a Linux HPC cluster
- Resolved correctness issues in real-time data pipelines by debugging and profiling memory bottlenecks and race conditions in multi-threaded C++ processing code
- Lowered the barrier for non-engineering researchers to run reproducible experiments independently by productionizing research code with versioning, testing, and CI/CD via GitHub Actions
- Selected as 1 of 15 from 250 applicants and received a return offer based on internship performance

### Kashmir World Foundation

Apr. 2024 – Jan. 2025

*Software Developer Intern*

*Ashburn, VA*

- Enabled field researchers across 12 countries to collect data in remote areas without network access by building a cross-platform React Native app in TypeScript with offline-first support
- Handled 150+ daily requests across mobile and web platforms by designing RESTful APIs with authentication, rate limiting, and structured data validation
- Surfaced population trends for conservation decision-making by developing React.js and D3.js dashboards that processed and visualized 500+ wildlife observations

## PROJECTS

### InterviewAI: AI Interview Simulation Platform | [GitHub](#) | [Video Demo](#)

Nov. 2025 – Present

- Built a multi-agent interview system using LangChain and GPT-4 that parses resume content to dynamically generate role-specific questions and adjusts difficulty based on response quality across multi-turn sessions
- Enabled context-aware follow-up questions and voice interview support by implementing agent memory, reasoning chains, and speech-to-text and text-to-speech API integrations
- Tracked user progress across sessions by deploying a full-stack solution on AWS (EC2, S3, IAM) with a FastAPI backend, Next.js frontend, and PostgreSQL database

## LEADERSHIP

### App Development Club (Founder) (2023–Present)

- Grew club to 20+ students by leading workshops on algorithm design, full-stack systems, and API development
- Designed and evaluated technical exercises to assess systems thinking and problem-solving under constraints

## TECHNICAL SKILLS

**Languages:** Python, Java, C/C++, SQL, JavaScript, TypeScript, Bash

**Cloud & Infrastructure:** AWS, Linux HPC clusters, Docker, CI/CD (GitHub Actions), PostgreSQL, Redis, MongoDB

**ML / AI:** PyTorch, CUDA, LangChain, OpenCV, NumPy, LLM evaluation, prompt engineering

**AI Dev Tools:** Claude Code, Codex, GitHub Copilot